Acute Gastrointestinal Bleed

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Upper Gastrointestinal Bleeding

Upper gastrointestinal (UGI) bleeding is proximal to the ligament of Treitz. UGI bleeding typically presents with hematemesis (vomiting of blood or coffee ground-like material) and/or melena (black and tarry stools).

Etiology

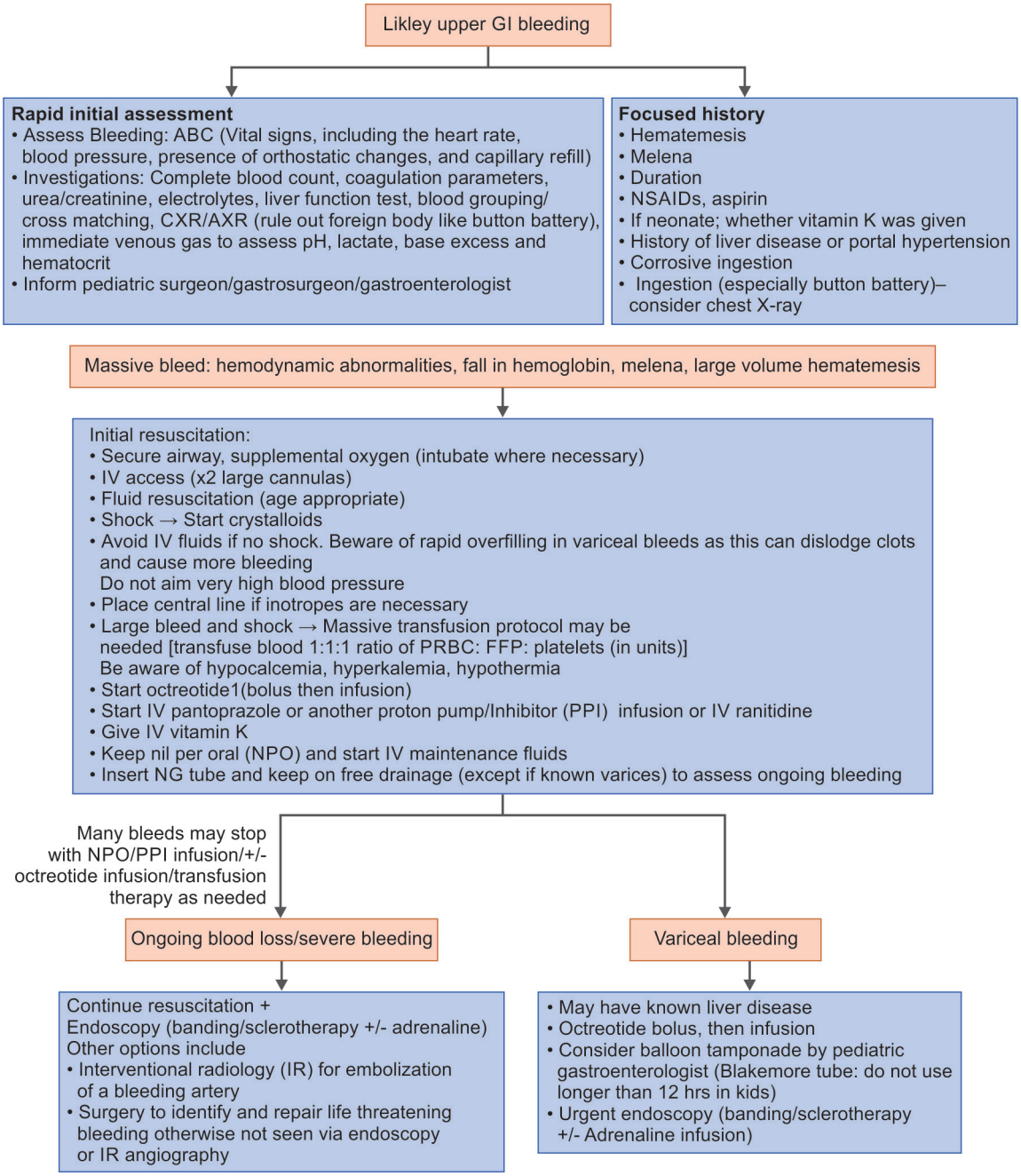
Table 1 presents etiology of UGI bleeding.

TABLE 1: Etiology of upper gastrointestinal bleeding.

Neonate	***Infant***	***Child or adolescent***
Swallowed maternal blood	Stress gastritis or ulcer	Mallory–Weiss tear (associated with vomiting)
Vitamin K-deficient bleeding	Acid peptic disease	☑ Acid peptic disease ☑ Drug induced, nonsteroidal anti-inflammatory drugs (NSAIDs)-induced ulcers
Stress gastritis or ulcer	Mallory–Weiss tear (associated with vomiting)	Gastric or esophageal varices
Esophagitis Trauma (e.g., nasogastric tube) Vascular anomalies	☑ Esophagitis ☑ Vascular anomalies ☑ Gastrointestinal duplications	☑ Esophagitis ☑ Foreign body ☑ Caustic ingestion
Gastrointestinal duplications	Gastric or esophageal varices	Vasculitis (e.g., Henoch–Schönlein purpura)
Coagulopathy (e.g., associated with infection)	Duodenal or gastric webs	Crohn's disease
Milk protein intolerance	Bowel obstruction	Bowel obstruction
Congenital coagulation factor deficiency	Milk protein allergy	Dieulafoy lesion
		Hemobilia

Flowchart 1 described management of UGI bleeding.

Flowchart 1: Management of upper gastrointestinal (UGI) bleeding.



(ABC: age, biomarkers, and clinical history; AXR: abdominal X-ray; CXR: chest X-ray; FFP: fresh frozen plasma; IV: intravenous; NSAIDs: nonsteroidal anti-inflammatory drugs; PRBC: packed red blood cell)

Doses

Octreotide

- ☑ 1–2 µg/kg IV bolus followed by:
  - Continuous infusion @1–2 µg/kg/hour (maximum 4 µg/kg/hour)
  - Diluent: 5% dextrose of 0.9% NaCL

Pantoprazole

- ☑ 5–15 kg: 2 mg/kg/dose IV and then 0.2 mg/kg/hour IV
- ☑ >15–40 kg: 1.8 mg/kg/dose IV and then 0.18 mg/kg/hour IV
- ☑ >40 kg: 80 mg/dose IV and then 8 mg/hour
- ☑ Maximum infusion duration: 72 hours

Further Reading

- ☑ Gilger MA. Upper gastrointestinal bleeding. In: Walker WA, Goulet O, Kleinman RE, Shneider BL, Sanderson IA (Eds). Pediatric Gastrointestinal Disease, 4th edition. Ontario: BC Decker Inc.; 2004. pp. 259-65.
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- ☑ Siafakas C, Fox VL, Nurko S. Use of octreotide for the treatment of severe gastrointestinal bleeding in children. J Pediatr Gastroenterol Nutr. 1998;26:356-9.

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